Unknown Motion Rejection in Myoelectric Pattern Recognition Using Convolutional Prototype Network

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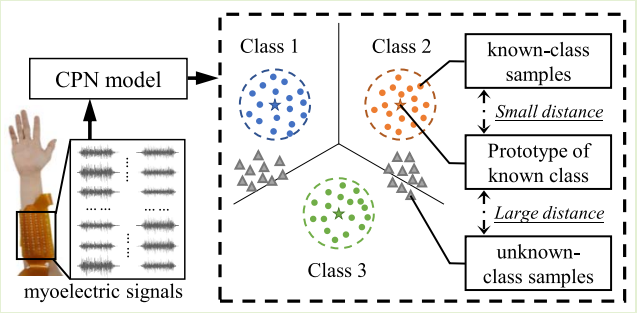
***Abstract*—Traditional myoelectric pattern recognition (MPR) systems usually perform well under laboratory conditions. However, in practice, unknown motions that do not appear in the training phase will inevitably disturb MPR systems, leading to dramatic performance degradation in motion classification. Generative and discriminative models are two representative ways to solve this problem in the literature. They only learn the patterns of known motions and reject unknown motion samples endowed with different patterns. However, the generative models often perform poorly in categorizing known motions compared with the discriminative models. While in contrast, the discriminative models always perform relatively inferiorly in distinguishing unknown motions. In this paper, a novel myoelectric control method based on the convolutional prototype network (CPN) is proposed. CPN combines the advantage of the generative model and the discriminative model in an end-to-end manner. It constructs multiple prototypes of known motion classes and identifies samples of known/unknown classes using the template (i.e., prototype) matching. The proposed method is evaluated using the high-density surface electromyography (HD-sEMG) data recorded while performing seven known and six unknown hand motions. Compared to the latest deep learning-based study, our proposed method can yield a significantly higher accuracy of 97.87% in identifying unknown motions and reduce training and testing time by more than 80%, indicating that it is a promising solution to realize robust myoelectric control in practice.**



***Index Terms*—Human-machine interface, myoelectric signal, pattern recognition, deep learning.**

I. Introduction

ELECTROMYOGRAPHY (EMG) signals are electrical biosignals that can reflect muscle contractions, therefore containing rich motor control information [1]. Surface EMG (sEMG) records activities of superficial muscles using electrodes placed on the skin surface over the tested muscles, which has the advantage of being non-invasive and easy to use. Over the past few decades, a large body of research has concentrated on the sEMG-based man-machine interface. In particular, sEMG is often chosen as the control source of hand prostheses [2]–[6], exoskeletons robots [7]–[9], or wearable devices [10]–[12].

Recent research efforts have developed the technology of myoelectric pattern recognition (MPR) that enables dexterous control of up to 20 activation patterns [13], [14]. Different motions resulting from different muscle activation patterns generate different sEMG patterns. MPR approaches aim to accurately chart these patterns/regularities from sEMG signals using computer algorithms, and with these regularities to interpret motion intentions. Typically, the combination of the time-domain (TD) features and the linear discriminant analysis (LDA) classifier [6], [15] is popular due to its superior performance and ease of implementation.

Despite the increase in efficiency, MPR systems tend to degrade performance under non-ideal conditions. Some issues, such as electrode shift [16]–[19], variation of contraction strength or limb position [3], [10], [20], [21], interference of unknown motions [22]–[32], may significantly impede the application of the MPR technology in real-world scenarios.

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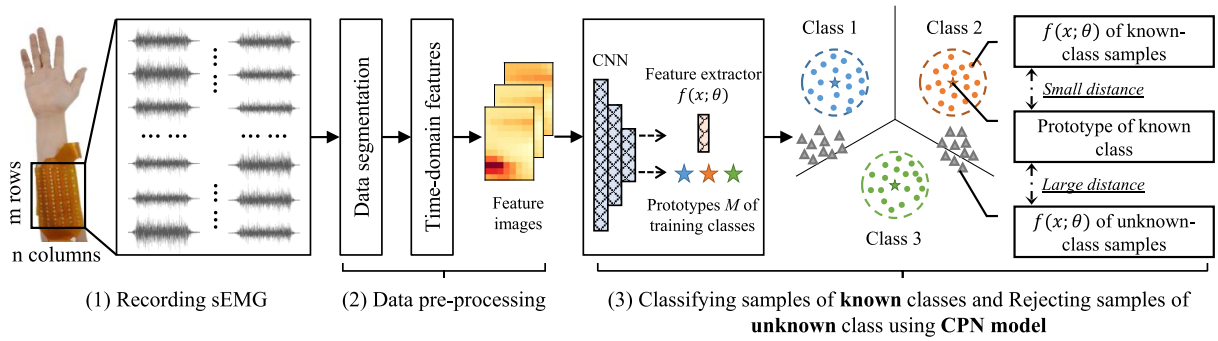


Fig. 1. Flowchart of the proposed method.

Among them, the unknown motion interference is a critical issue.

Traditional MPR approaches often evaluate performances under the closed-set setting, which means that MPR approaches only define limited known classes during training and testing [6], [13], [17]. That is to say, the users can only interact with the environment using the limited set of motions the MPR system can respond to. However, in real-world scenarios, unknown motions, i.e., motions not appearing in training, usually arise in the testing stage [30]. An unknown motion leading to unseen patterns will confuse the MPR system, and the system is very likely to misclassify the unknown motion into one of the known motion classes [27], [32]. Actually, it is hard to limit the users to merely perform a specific set of known motions during activities of daily living. At the same time, sEMG is very sensitive to muscle activation, and some unconscious muscle contractions during motion transitions or changes in limb positions can also introduce unknown motion interference. As a result, unsmooth control caused by frequent erroneousness will provide negative feedback to users [31]. Therefore, it is essential to enable MPR systems to distinguish and reject unknown motions.

Many studies have been conducted to distinguish unknown motions [22]–[32]. The usual solutions are to model sEMG patterns of all known motions and subsequently reject unknown class samples that significantly different from known patterns. Therefore, the main challenges lie in accurately charting the intrinsic regularities of known motions and finding a proper similarity function to distinguish unknown class samples. A typical solution utilizes classification probabilities output by discriminative models, such as support vector machine (SVM) [22], convolutional neural network (CNN) [23], to reject unknown class samples. These approaches expect that discriminative models can model known patterns well, and unknown class samples are endowed with relatively low probabilities. Although discriminative models often achieve high classification accuracies on known motion classes, they usually allocate the unknown class sample to a known class with improperly high probability due to the closed-set setting during training [33]. Take CNN as an example; this model assigns the whole feature space to the known classes by optimizing the softmax loss. In this way, an unknown sample will inevitably misclassfied into a known class, and the model is very likely to output an incorrectly high probability [34].

To further improve the performance of detecting unknown motions, some researchers attempt to adopt generative models, which purely model the probability distribution of each known class. If the likelihoods of the testing sample estimated from all known class models are low, the sample will be assigned to the unknown class. Typically, LDA [24], Mahalanobis distance [25]–[27], support vector data description (SVDD) [28], [29], multiclass one-versus-one classifiers [30], [31] and autoencoder [32] are popular approaches applied in the field of myoelectric control. However, compared with the aforementioned solutions based on discriminative models, the difficulty of accurately modeling known patterns often make the generative models achieve inferior classification performance on known patterns.

An ideal solution should be able to accurately categorize known motions and detect unknown motions simultaneously. To this end, Wu *et al.* [32] combine a classifier and multiple generative models in a deep-learning way. They suppose that the classifier can improve the classification performance on known patterns, and the generative models are able to detect unknown patterns effectively. However, multiple generative models inevitably increased computational complexity. Besides, the separate training processes of the classifier and generative models limits the overall performance. To cope with this disadvantage, inspired by Yang *et al.* [34], [35], we introduce a novel myoelectric control method based on convolutional prototype network (CPN). The CPN aims to construct a CNN feature extractor and multiple prototypes of known classes. It simulates the recognition mechanism of humans, where the humans extract abstract representations (i.e., feature extractor) from objects and identify unknowns using template (i.e., prototypes) matching. Compared with our previous work [32], CPN combines the advantages of the generative model and the discriminative model in an end-to-end manner, which can achieve superior performance of unknowns detection, with a lower computational burden. In the following chapters, we will introduce the details of the proposed method.

II. Method

The flowchart of the proposed method is shown in Fig. 1. The recorded sEMG signals of different known motions are first processed into feature images. Then, these images, with corresponding motion labels, are fed to the CPN model to obtain a feature extractor and multiple prototypes. In the

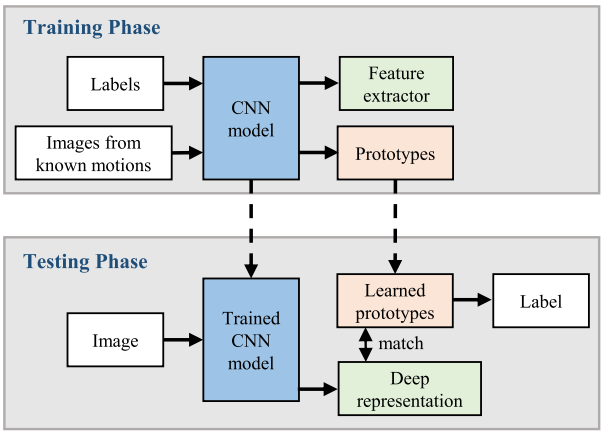


Fig. 2. The training and testing phases of the CPN method.

testing phase, the sample close to any prototype is assigned to one known class, whereas the sample far from all prototypes is called the unknown class sample.

A. sEMG Recording and Pre-Processing

In this study, High-density sEMG (HD-sEMG) is adopted to collect sufficient motor control information. Compared with traditional bipolar electrodes located on different muscles, HD-sEMG increases the spatial resolution and electrode number, using arrays of closely spaced electrodes. The additional spatial-temporal information derived from HD-sEMG has been proven to improve characterization ability of muscle activation patterns [17], [18], [27], [32], [36], which is important in detecting unknown motion.

As shown in Fig. 1(1-2), the recorded HD-sEMG signals are first segmented into a succession of 250ms analysis windows with an overlap of 150ms. Then, the windows of the no motion class are discarded using the amplitude threshold manner [24]. For each analysis window of active motion classes, we extract three time-domain features [6], namely root mean square (RMS), mean absolute value (MAV) and waveform length (WL), from each channel. Therefore, suppose that the channel deployment of the adopted HD-sEMG electrode array is $m \times n$, one analysis window is processed into a matrix in the form of $m \times n \times 3$, where 3 represents the three time-domain features. We call this processed matrix as a feature image (hereinafter referred to as "image"). The processing method described above is consistent with our previous work [32], and its effectiveness has been confirmed.

Accordingly, the images of active motion class constitute a known-class dataset X that includes K samples from l known classes: $X = \{X_i, i = 1, 2, \ldots, l\} = \{(\boldsymbol{x}_k, y_k), k = 1, 2, \ldots, K\}$, where $\boldsymbol{x}_k \in \mathbb{R}^{m \times n \times 3}$ and y_k labels the sample $\boldsymbol{x}_k$ into one of the l known classes.

B. Constructing Convolutional Prototype Network (CPN)

Fig. 2 demonstrates the training and testing phases of the CPN method. In the training phase, CPN aims to learn a feature extractor $f(\boldsymbol{x}; \theta)$ and a prototype set $M = \{\boldsymbol{m}_i\}$ simultaneously [34] from the known class dataset, where $i \in \{1, 2, \ldots, l\}$, $\boldsymbol{x}$ refers to an inputted image, and θ represents parameters of the network. In the testing phase, an image $\boldsymbol{x}$ belongs to class i only when its deep representation $f(\boldsymbol{x}; \theta)$ is close to the corresponding prototype $\boldsymbol{m}_i$. And if the $f(\boldsymbol{x}; \theta)$ is far from all prototypes in M, we call the $\boldsymbol{x}$ an unknown class sample (see Fig. 1(3)). Therefore, the probability of a sample $(\boldsymbol{x}, y)$ belonging to the prototype $\boldsymbol{m}_i$ can be defined as:

$$p(\boldsymbol{x} \in \boldsymbol{m}_i | \boldsymbol{x}) \propto -D(f(\boldsymbol{x}; \theta), \boldsymbol{m}_i) \tag{1}$$

where the function $D(\cdot)$ measures the squared Euclidean distance between two elements:

$$D(f(\boldsymbol{x}; \theta), \boldsymbol{m}_i) = \| f(\boldsymbol{x}; \theta) - \boldsymbol{m}_i \|_2^2 \tag{2}$$

Next, the probability $p(\boldsymbol{x} \in \boldsymbol{m}_i | \boldsymbol{x})$ is calculated as:

$$p(\boldsymbol{x} \in \boldsymbol{m}_i | \boldsymbol{x}) = \frac{e^{-D(f(\boldsymbol{x};\theta), \boldsymbol{m}_i)}}{\sum_{c=1}^{l} e^{-D(f(\boldsymbol{x};\theta), \boldsymbol{m}_c)}} \tag{3}$$

The Eq. (3) makes the CPN able to categorize samples according to their distances from prototypes M. Then, the classification probability that the sample $\boldsymbol{x}$ belongs to the label y is defined as:

$$p(y|x) = p(\boldsymbol{x} \in \boldsymbol{m}_y | \boldsymbol{x}) \tag{4}$$

Later, Cross-entropy (CE) loss is adopted to optimize the parameters of the network:

$$\mathcal{L}_{ce}((\boldsymbol{x}, y); \theta, M) = -\ln(p(y|x)) \tag{5}$$

Optimizing the CE loss makes each sample $\boldsymbol{x}$ close to its corresponding prototype $\boldsymbol{m}_y$ and, simultaneously, stay away from other prototypes. However, CE loss still assign the whole feature space to known classes. For example, the decision boundary between two adjacent classes is:

$$\| f(\boldsymbol{x}; \theta) - \boldsymbol{m}_i \|_2^2 = \| f(\boldsymbol{x}; \theta) - \boldsymbol{m}_j \|_2^2 \tag{6}$$

then

$$f(\boldsymbol{x}; \theta) \cdot \underbrace{2(\boldsymbol{m}_j - \boldsymbol{m}_i)}_{\text{weight}} + \underbrace{\|\boldsymbol{m}_i\|_2^2 - \|\boldsymbol{m}_j\|_2^2}_{\text{offset}} = 0 \tag{7}$$

thus, the whole feature space is linearly divided into l sub-regions, leaving no space to unknown classes and inevitably leading to poor performance on rejecting unknown class samples. To this end, an extended loss, prototype loss (PL), is added to constraint the sample distributions of known classes:

$$\mathcal{L}_{pl}((\boldsymbol{x}, y); \theta, M) = \| f(\boldsymbol{x}; \theta) - \boldsymbol{m}_y \|_2^2 \tag{8}$$

where $\boldsymbol{m}_y$ is the prototype of sample $\boldsymbol{x}$ belongs to. The PL can make intra-class distribution more compact. Finally, a joint loss is constructed by combining the CE loss and the PL:

$$\begin{aligned} &\mathcal{L}_{total}((\boldsymbol{x}, y); \theta, M) \\ &\quad = \mathcal{L}_{ce}((\boldsymbol{x}, y); \theta, M) + \lambda \mathcal{L}_{pl}((\boldsymbol{x}, y); \theta, M) \end{aligned} \tag{9}$$

where λ balances the two losses. With this joint loss, $f(\boldsymbol{x}; \theta)$ can characterize representations of images in both the compact intra-class variations (using PL) and separable inter-class differences (using CE loss), thus integrating the generative model and the discriminative model in an end-to-end way.

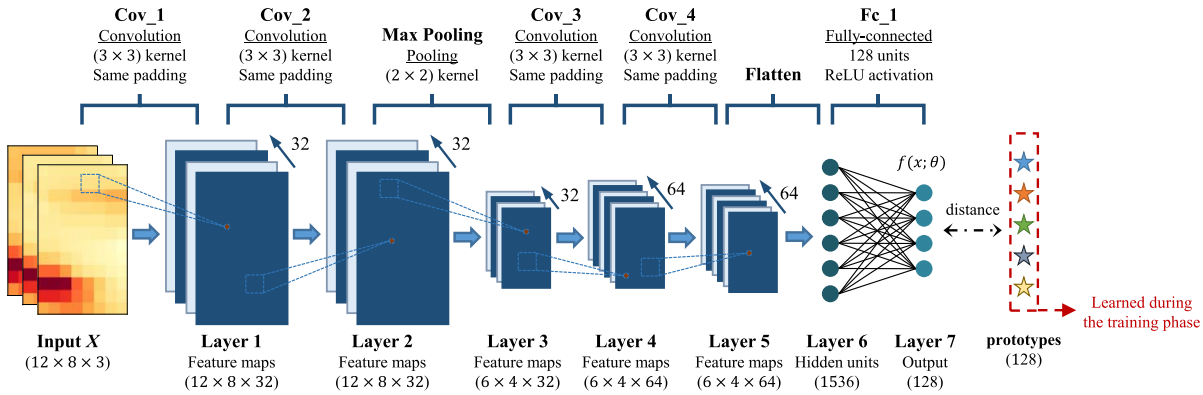


Fig. 3. The architecture of the CPN adopted in this study. This network consists of four convolutional layers, one fully connected layer. The detailed parameter settings of each layer are listed in the upper part of the figure.

The Fig. 3 demonstrates the CPN configuration adopted in this study. Multiple convolutional layers are utilized to extract spatial information from HD-sEMG images, and a fully-connected layer is added to the last to produce the deep representation $f(\boldsymbol{x};\theta)$ of the inputted sample $\boldsymbol{x}$. We select this network architecture after some tests. Combined with the joint loss $\mathcal{L}_{total}$, we call for the Adam [37] algorithm to optimize the network parameters θ and the prototypes M. The batch size is set to 50. We train the network for 200 epochs. The learning rate is initialized as 0.001 and dropped by half every 50 epochs. The $L2$ regularization with the weight of $1e-6$ is added to all parameters to prevent overfitting.

C. Class Prediction Using CPN

In the testing phase, the predicted label y_p of the inputted sample $\boldsymbol{x}$ is calculated through:

$$y_p = \arg\max_{i=1}^{l} p_t(\boldsymbol{x} \in \boldsymbol{m}_i|\boldsymbol{x}) \tag{10}$$

where

$$p_t(\boldsymbol{x} \in \boldsymbol{m}_i|\boldsymbol{x}) = \frac{e^{-\|f(\boldsymbol{x};\theta)-\boldsymbol{m}_i\|_2}}{\sum_{c=1}^{l} e^{-\|f(\boldsymbol{x};\theta)-\boldsymbol{m}_c\|_2}} \tag{11}$$

Here, the Euclidean distance that possesses a relatively steep graph is used to measure the distance between the representations and prototypes. We found this distance is sensitive to differences between samples. Next, a threshold T need to be determined to identify unknown motions:

$$y_{novel} = \begin{cases} 0 & \text{if } p_t(\boldsymbol{x} \in \boldsymbol{m}_{y_p}|\boldsymbol{x}) > T \\ 1 & \text{if } p_t(\boldsymbol{x} \in \boldsymbol{m}_{y_p}|\boldsymbol{x}) \leq T \end{cases} \tag{12}$$

Therefore, when sample $\boldsymbol{x}$ is inputted to the CPN, we first calculated its maximal predicted probability $p_t(\boldsymbol{x} \in \boldsymbol{m}_{y_p}|\boldsymbol{x})$. A threshold is then used to determine whether the sample is from the unknown class ($y_{novel} = 1$). If the sample is from known class ($y_{novel} = 0$), its predicted label is y_p.

We implemented all algorithms mentioned above using Python language and TensorFlow architecture, and ran them on a desktop PC equipped with an Intel i9 CPU and an NVIDIA GeForce GTX 2080Ti GPU.

D. Evaluation Metrics

Two metrics are applied to evaluate the detection performance of unknown motions and the classification performance of different motions, respectively. They are receiver operating characteristic (ROC) curve [38] and the classification accuracy.

ROC curve reflects the overall diagnostic ability of a binary classifier. This curve plots the true positive rate (TPR) against the false positive rate (FPR) under different threshold (i.e., T in this study) settings. Specifically, the TPR measures the proportion of known motion samples that are correctly identified. The FPR, also known as false alarm ratio, measures the proportion of unknown motion samples that are wrongly classified as the known motion class. Ideally, 100% TPR combined with 0% FPR is a perfect situation. The area under the curve (AUC) summarizes the ROC curve in a single number, where a higher AUC always indicates a better performance of the binary classifier.

Next, when a threshold is determined, we also calculate the classification accuracy of each motion class. The classification accuracy is the proportion of samples that are correctly identified. The proposed method is evaluated in a user-specific manner, and all metrics of each subject are averaged from five repetitions of the experiments.

E. Comparison Methods

Three baseline methods are implemented to evaluate the effectiveness of the proposed method:

LDA [6]. This method first extracts four time-domain features from each signal window. Then the LDA classifier is used to categorize samples. The LDA method is widely adopted as the baseline method in the myoelectric control field.

LDA with Mahalanobis distance (LDA-MD) [25], [26]. This method is an upgraded version of the previous method. The additional MD measures the distance between the testing sample and the distributions of the training data:

$$MD_{\boldsymbol{x}} = (\boldsymbol{e}_{\boldsymbol{x}} - \boldsymbol{\mu}_{y_x})^T \Sigma_{y_x}^{-1} (\boldsymbol{e}_{\boldsymbol{x}} - \boldsymbol{\mu}_{y_x}) \tag{13}$$

where $\boldsymbol{x}$ is the testing sample and $\boldsymbol{e}_{\boldsymbol{x}}$ is the feature vector after dimensionality reduction by LDA. y_x is the predicted label by LDA. $\boldsymbol{\mu}_{y_x}$ and Σ_{y_x} are class mean vector and covariance matrix of the training data of class y_x, respectively. The MD

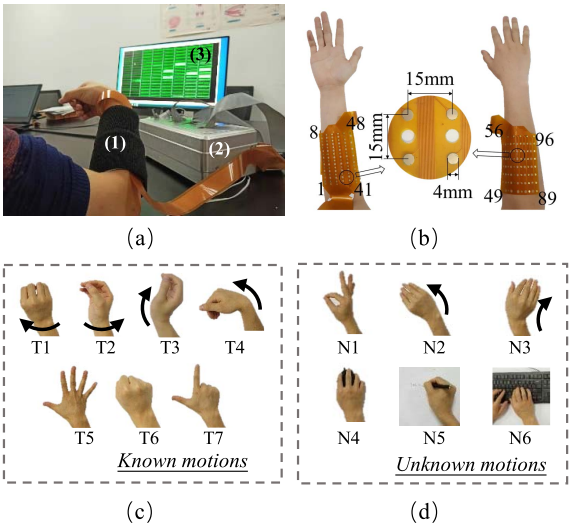


Fig. 4. Experimental device and motions. **(a)** shows the custom HD-sEMG acquisition system, including two pieces of high-density electrode array (1), signal recording device (2), and a waveform display software (3). **(b)** shows the placement and the detailed configuration of two electrode arrays. **(c)**, **(d)** demonstrate the known and unknown motions adopted in this study.

utilizes $\boldsymbol{\mu}$ and Σ to construct distributions/prototypes of known motion classes. If the testing sample is far away from all known distributions, we call it the unknown class sample.

CNNSC+AE [32] method first utilizes CNN to extract discriminative representations of the HD-sEMG images, and then rejects unknown class samples through autoencoders. This method has reported the best performance in unknown motion rejection. We implemented all the above comparison methods following the settings in [32].

F. Statistics Analysis

One-way repeated measures ANOVAs were used to compare the classification performances from four methods (LDA, LDA-MD, CNNSC+AE and the CPN method). Post hoc tests were conducted if necessary. The significant level was set to 0.05. All these analyses were performed using SPSS software (ver.24.0, SPSS Inc. Chicago, IL, USA).

III. Experiments and Results

A. Experiments Setup

1) Subjects: We recruited nine intact subjects (25.33 ± 1.56 years old; six males, three females) to participant in the experiments. The Ethics Review Board of the University of Science and Technology of China has approved the data collection experiments. The aims and content of the experiment were detailed introduced to each subject, and all subjects have signed written consents before experiments.

2) Data Acquisition System: A custom HD-sEMG acquisition equipment was adopted in this experiment (see Fig. 4a-b), constituting two high-density electrode arrays, one signal acquisition device, and the display software. The electrode arrays that surround the forearm detect sEMG from extensors and flexors. Each array contains 48 monopolar electrodes arranged in the form of an 8×6 matrix. The inter-electrode distance and electrode diameter are 15mm and 4mm, respectively.

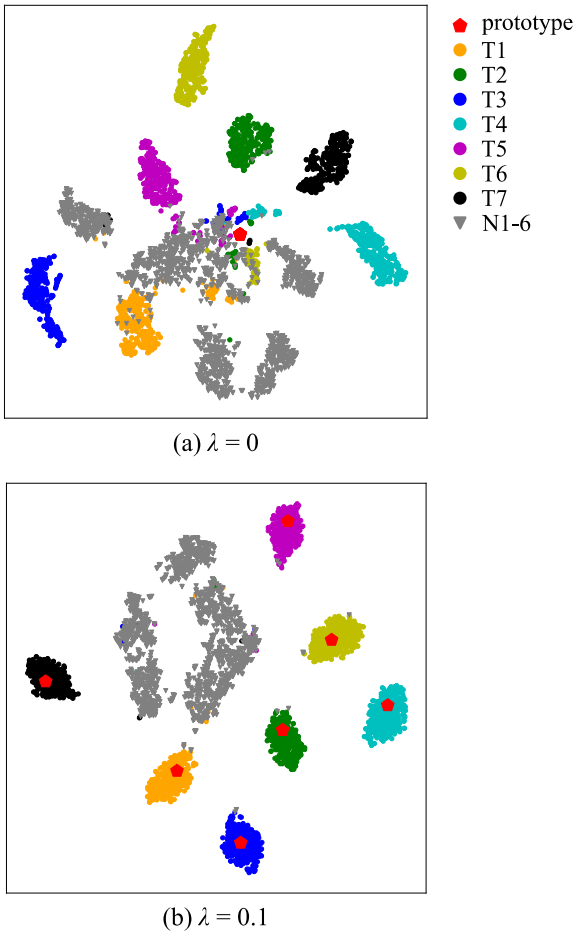


Fig. 5. t-SNE visualization of deep features trained with different hyper-parameter λ. The scattered points with different shapes indicate prototypes or samples of different motion classes.

The detected sEMG signals are then amplified with a gain of 60dB and filtered to 20-500Hz by the recording device. Next, A 16-bit analog-to-digital converter is used and the sampling rate is 1 kHz. The signal display software finally visualizes and saves the digital signals.

3) Data Collection Procedures: During the experiment, the subjects were asked to sit comfortably in a chair, and their forearm limbs were relaxed on a table. Before placing the electrode arrays, the corresponding skin areas were cleaned by rubbing alcohol. A stretchable textile was used to make the electrode-skin contact more stable.

Thirteen motions, including seven known/target motions (Fig. 4c) and six unknown/novel motions (Fig. 4d), were investigated in this study. The known motions were traditional human-machine interaction gestures that were widely studied in [6], [17], [36], and they were wrist pronation (T1)/supination (T2), wrist extension (T3)/flexion (T4), hand open (T5)/close (T6) and shoot (T7). The unknown motions can be divided into static and dynamic types. Static motions were isometric contractions that were obviously different from the known motions, and we collected three static motions: pinch (N1), radial deviation (N2), and ulnar deviation (N3). This type of unknown motion was always investigated in previous works [23], [25], [26], [30]. However, in practice, users

TABLE I
DATA USED IN THIS STUDY

Subject	Gender	Age	Training set (known motions)	Testing set (known motions)	Testing set (unknown motions)
1	M	26	510.31s	127.58s	469.95s
2	M	25	551.95s	137.99s	459.30s
3	M	27	588.67s	147.17s	484.20s
4	F	24	488.95s	122.24s	522.30s
5	M	27	298.03s	74.51s	338.55s
6	F	22	566.35s	141.59s	456.00s
7	M	25	411.80s	102.95s	388.65s
8	M	27	311.60s	77.90s	392.85s
9	F	25	417.91s	104.48s	382.50s
Average	n.a.	25.33±1.56	460.61±101.12s	115.15±25.28s	432.70±56.01s

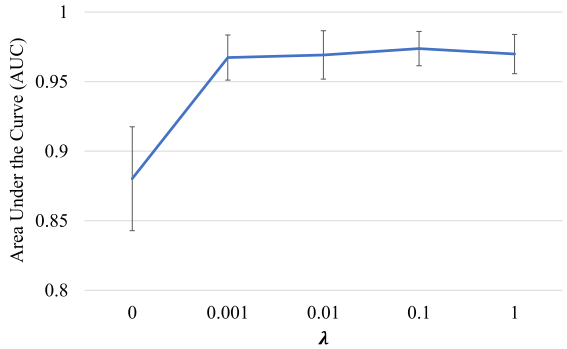


Fig. 6. AUC values achieved by the CPN using different λ. Each point is average from all subjects. Error bars represent standard errors.

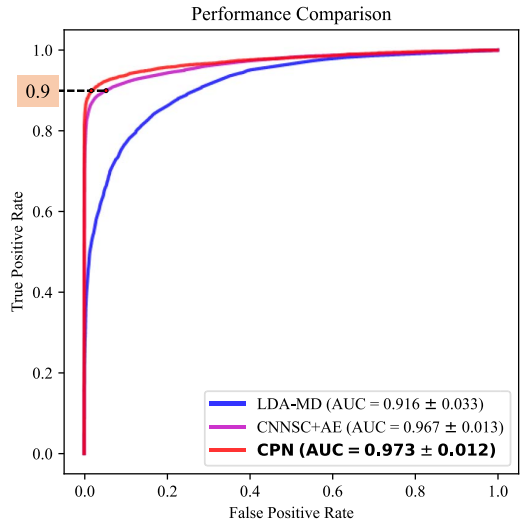


Fig. 7. ROC curves averaged over all subject of three methods.

seldom perform an apparent unknown gesture on purpose. More unknown class samples could occur during performing daily tasks such as mouse manipulating (N4), handwriting (N5), and keyboard typing (N6). We selected these motions because they often contained a single or combination of the adopted known motions, which increased the rejection difficulty. For T1-7 and N1-3 motions, data were collected five trials, each containing two 5-s repetitions of each motion class. For each motion of N4-6, subjects were instructed to perform for roughly 1 minute while the sEMG data were recorded. Sufficient rest between two repetitions was given to avoid fatigue.

Data of known motion classes were further split into training and testing set, following the ratio of 80%, 20%. Besides, all data of unknown motion classes were added to testing set. The detailed descriptions of the data are demonstrated in Table I.

B. Performance on Unknown Motion Rejection

As in section II-B and II-C, two hyper-parameters, i.e., λ in Eq. 9 and T in Eq. 12, may affect the performance of unknown motion rejection.

The λ controls the PL weight in the joint loss, which dominants the compactness of the deep features. Fig. 5 shows the visualizations of deep features from different λ using t-distributed stochastic neighbor embedding (t-SNE) technology [39]. With a proper λ, the deep features within the same class are more compact. Furthermore, the samples from known and unknown classes are less overlapped in the feature space. Fig. 6 shows the AUC values obtained from the CPN under different λ settings. A higher AUC value indicates a better ability to detect unknown motions. Note that, a large range of λ can improve the CPN's performance. Especially, the AUC value reaches the maximum of 0.973 when the λ is 0.1. In the following analyses, we set λ to be 0.1.

Fig. 7 shows the average ROC curves of the three methods using all testing data from all subjects. The LDA method is neglected because it cannot reject any unknown motion. These plots summarize the trade-off between the TPR (sensitivity) and FPR (false alarm ratio) using different probability thresholds, and the most desirable ROC curve is the one approaching the top left quadrant. The CPN yields the best ROC curve that achieves the most considerable AUC value of 0.973. Furthermore, it should be noted that there is an apparent difference in FPR when the TPR is reaching 0.9, which means that the CPN performs better than the other classifiers under a strict setting.

Fig. 8 shows the CPN's performance under different probability thresholds T. This plot indicates that the thresholds between 0.64 and 0.76 are reasonable because they make the classification accuracies of known and unknown classes both higher than 90%. Table II summarizes the classification performance of four methods, averaged across all subjects and testing sets when a proper threshold is selected for each method (except for the LDA method). For the CPN case, the threshold T is set to 0.76. Although the LDA method achieves better performances on known motion classification, it cannot reject any unknown motion. All methods can achieve about 90% accuracy on known motions classification. Furthermore,

TABLE II
CLASSIFICATION ACCURACIES (%) OF KNOWN AND UNKNOWN CLASSES USING FOUR METHODS. FOR EACH METHOD (EXCEPT FOR LDA), A PROPER THRESHOLD IS SELECTED TO MAKE THE AVERAGE ACCURACY OF KNOWN CLASSES COMPARABLE. ALL ACCURACIES ARE AVERAGED ACROSS ALL SUBJECTS

Method	Known motions	Unknown motions						
	Average	typing	writing	mouse manipulating	pinch	ulnar deviation	radial deviation	Average
LDA [6]	**95.01±1.99**	0	0	0	0	0	0	0
LDA-MD [25], [26]	88.43±2.73	82.30±11.72	75.52±16.30	79.93±11.41	53.85±22.66	83.96±14.44	89.37±9.26	77.49±18.78
CNNSC+AE [32]	88.60±2.26	96.44±3.98	96.67±4.31	94.70±6.21	96.22±3.20	90.07±20.10	91.59±11.16	94.28±10.42
CPN	89.66±3.48	**99.11±0.89**	**98.26±3.20**	**98.93±1.16**	**97.33±4.75**	**97.44±3.31**	**96.15±3.91**	**97.87±1.45***

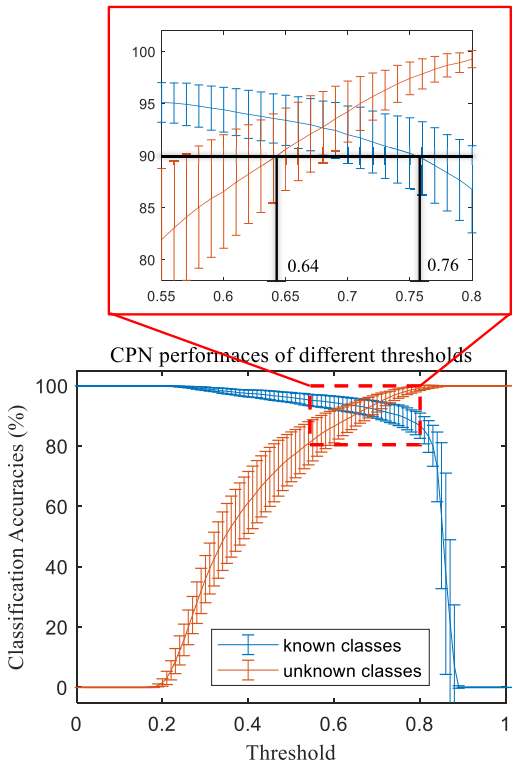


Fig. 8. CPN performance on identifying known and unknown motion samples with different thresholds. The accuracies are averages over all subjects. Error bars represent standard errors.

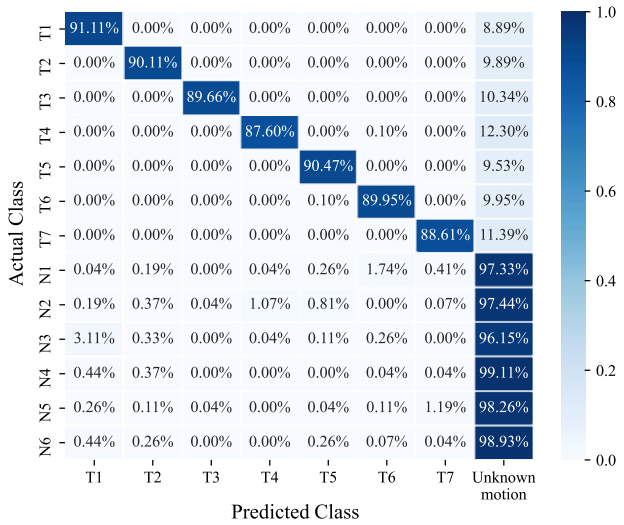


Fig. 9. Confusion matrix of CPN, averaged across all subjects and testing sets. T1-7 denote seven known motion classes, while N1-6 represent six unknown motion classes. The threshold T is set to 0.76.

the CPN scheme produces 97.87% accuracy on identifying unknown motion classes, which is significantly better ($p < 0.001$) than the latest CNNSC+SC method and achieves state-of-the-art performance. Furthermore, Fig. 9 shows the corresponding confusion matrix of CPN. It should be noted that, although a portion of known class samples are misclassified, they are rejected and predicted as the unknown class, which will not lead to apparent error response of the system. Fig. 10 demonstrates an example of online results produced by CPN. Note that, most misclassified known class samples are concentrated in the transient phases of the contractions, while most unknown class samples are correctly recognized.

C. Experiments for Routine Myoelectric Control

In this section, the CPN method is compared with the other three methods under the routine myoelectric control scenario. That is to say, the training data and the testing data only come from known motion classes. The classification performances of all methods are listed in Table III. In this case, the CPN method achieves significantly better classification accuracies for most subjects ($p < 0.05$).

TABLE III
AVERAGE ACCURACIES (%) ON SEVEN KNOWN MOTION CLASSES

Subject	LDA [6] and LDA-MD [25], [26]	CNNSC+AE [32]	CPN
1	97.27	98.85	**99.14**
2	96.98	**98.99**	98.79
3	97.79	98.40	**98.40**
4	94.90	97.08	**97.96**
5	94.22	96.34	**97.11**
6	92.82	96.47	**97.77**
7	91.84	95.43	**96.52**
8	95.92	99.08	**99.39**
9	93.32	94.70	**97.77**
Average	95.01±1.99	97.26±1.55	**98.09±0.88**

TABLE IV
AVERAGE PROCESSING TIMES (S) OF THE FOUR METHODS

	Method	Training time	Testing time
Traditional	LDA [6]	**0.13**	$\mathbf{1.50}\times10^{-6}$
	LDA-MD [25], [26]	0.14	6.10×10^{-5}
Deep Learning	CNNSC+AE [32]	414.64	4.76×10^{-3}
	CPN	**29.47**	$\mathbf{7.25}\times10^{-4}$

D. Computation Time

The average training and testing times of four methods, i.e., LDA, LDA-MD, CNNSC+AE, and CPN, are listed

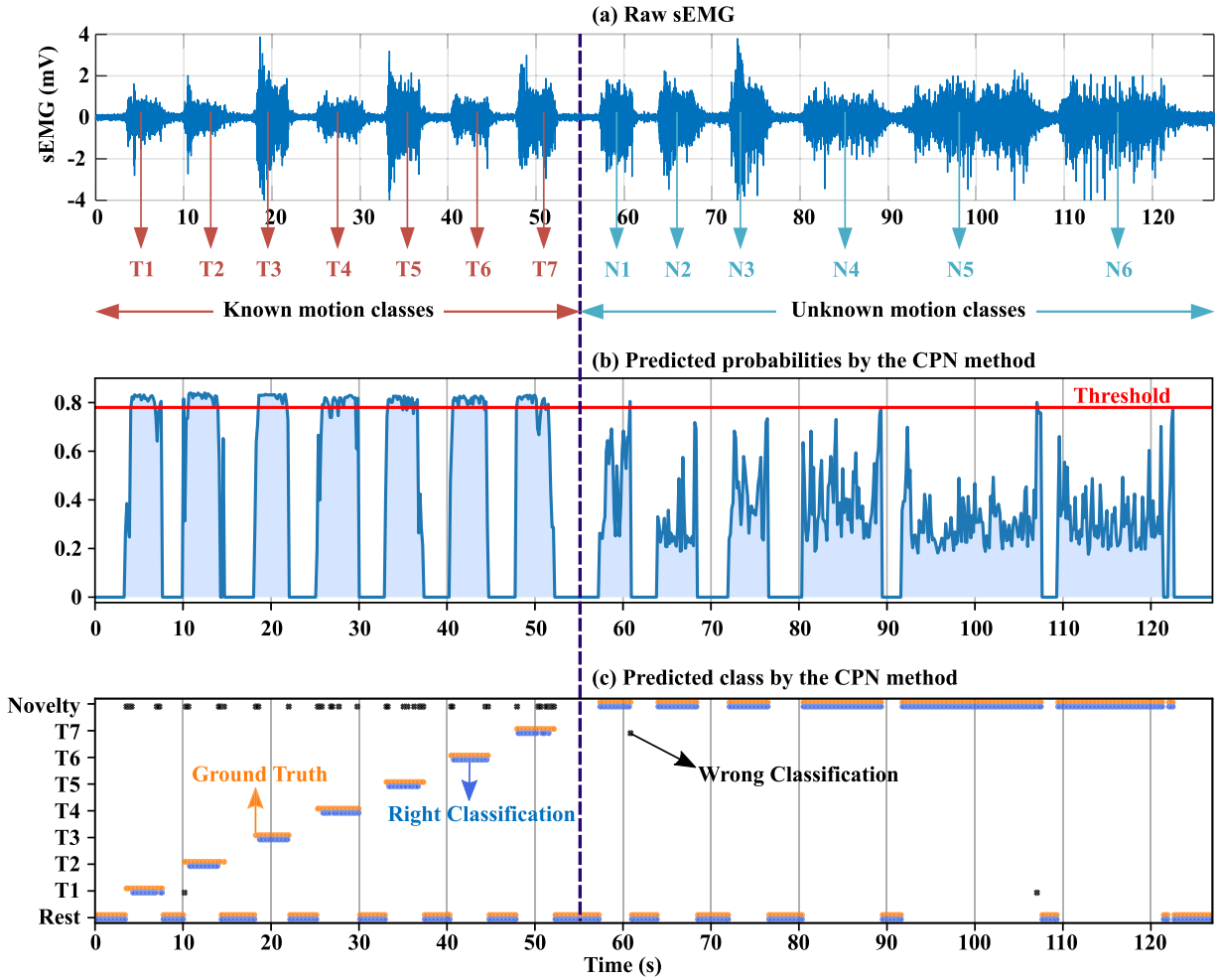


Fig. 10. Raw sEMG data of one channel (a) and the corresponding predicted probabilities (b) and classes (c) by the CPN. The data come from subject 1.

in Table IV. The traditional methods run faster than the deep learning methods. Besides, the CPN scheme is significantly faster than the CNNSC+AE ($p < 0.001$), where the CPN yields a 92% and 84% reduction in training and testing time, respectively.

IV. Discussion

It is crucial to reject unknown motions in robust myoelectric control. This paper proposed a rejection-capable CPN control scheme that can detect unseen/unknown muscular contraction patterns effectively.

Routine myoelectric pattern recognition schemes often separate the whole feature space into multiple sub-regions, and each region is assigned to one muscular contraction pattern. This strategy leaves no more room for unknown/unseen patterns and can't reject any unknown motions theoretically. As in Fig. 5, when the λ is set to 0, the CPN model that only trained with CE loss separates the whole feature space, and the unknown and known motion samples are mixed seriously, which hinders the rejection procedure. However, when a proper λ is selected (such as $\lambda = 0.1$), the deep features appear to be inter-class separable and intra-class compressed. It is noteworthy that, almost all samples of known motion classes are close to their corresponding prototypes, while the unknown motion samples are far from them. Pursuant to these visualizations, the discriminative deep features are hence more suitable for unknown motion rejection. Fig. 6 also approved the above presumption, in which a large range of λ can significantly improve the ability to identify unknown motion samples.

The probability threshold T controls the trade-off between the classification performance of known classes and unknown classes. Fig. 8 shows that, for all subjects, the CPN with a T range of 0.64 to 0.76 can ensure that the accuracies of detecting known and unknown motion samples both exceed 90%. This large range can be attributed to the relatively large distance between the unknown motion samples and all prototypes of known motion classes. As in Fig. 10b, the probabilities of unknown motion samples are apparently lower than that of known motion samples, making the rejection process easier.

However, although most unknown motion samples are correctly identified and rejected, a small portion of known motion samples are misclassified. According to the confusion matrix (Fig. 7) and the online results (Fig. 10), these wrong predictions often occur at the early and end phases

of the contractions, and most samples are misclassified as the unknown motion class. The transient state during switching between classes emanates from a burst of fibers, yielding ambiguous contraction patterns. In addition, compared with data from steady-state contractions, these transient-state data are always lesser, hence difficult in describing their intrinsic patterns [15]. Given that the CPN is sensitive to differences from known motion patterns, these ambiguous samples have a high probability of misclassification. However, due to the inertia of the devices, users may not feel the intermittent rejections during the control process. They will only notice that the control speed is occasionally slow [24], which can be improved by adding a proportional control strategy into the system. Besides, both dynamic and static types of unknown motion classes are investigated in this study. Note that previous studies always define static motions as unknown classes [25], [28]–[30], which reduces the detection difficulty because the distributions of static motions are clearly distinct from that of the known motions. The dynamic motions with arbitrary contractions accord with the practical applications and may introduce similar contractions with the known motions, making the rejection task more challenging. Therefore, the average accuracy of 97.87% in identifying unknown motions further validates the effectiveness of the proposed method.

Compared with the latest work [32] that extracted deep features and identified unknown classes in two separated steps, the CPN combined the two steps into a whole using prototype learning. In such a way, this end-to-end CPN model was able to: (1) detect unknown motion classes more accurately (Table II); (2) consist of fewer parameters and run more efficiently (Table IV). Furthermore, we also compare the CPN with other methods under a routine myoelectric control setting. That is to say, the motion classes in the testing set are consistent with those in the training classes. As in Table III, CPN achieves 98.09% accuracy, which is significantly better than that of the LDA-based methods ($p < 0.001$) and the CNNSC+AE method ($p < 0.05$). The improvement can be attributed to the powerful ability in non-linearly representation of the deep learning method. Moreover, given the training data are relatively small in myoelectric control field, it is suitable to apply prototype learning because of its advantages in dealing with the problem of small sample size [34].

Finally, there are also limitations in this study. We set only one threshold T for all classes. Ongoing work is to find class-specific thresholds for more accurate rejection. In addition, we concentrate only on detecting unknown motions. In practice, an ideal control scheme should be able to decode the user's contraction intentions, regardless of whether these intentions belong to the unknown classes. Current work is looking for a novel myoelectric control framework, such as modeling movements of individual fingers [40]. It is hoped that dexterous control can be realized through refined modeling of EMG. All these topics will be the directions of our future works.

V. Conclusion

This study proposed a rejection-capable myoelectric control scheme for alleviating the interference of unknown motions. The proposed method first converted HD-sEMG into feature images. Then a convolutional prototype network was adopted to learn prototypes of known motion classes. A sample was labeled as unknown if it was far from all prototypes. The proposed method achieved 97.87% accuracy on identifying six static and dynamic unknown motions, which was superior to other comparison methods. Our proposed method is a promising solution for alleviating interference of unknown motions in myoelectric control.

References

[1] C. J. De Luca, "The use of surface electromyography in biomechanics," *J. Appl. Biomech.*, vol. 13, no. 2, pp. 135–163, May 1997.

[2] J. O. de Oliveira de Souza, M. D. Bloedow, F. C. Rubo, R. M. de Figueiredo, G. Pessin, and S. J. Rigo, "Investigation of different approaches to real-time control of prosthetic hands with electromyography signals," *IEEE Sensors J.*, vol. 21, no. 18, pp. 20674–20684, Sep. 2021.

[3] R. Rajapriya, K. Rajeswari, D. Joshi, and S. J. Thiruvengadam, "Forearm orientation and contraction force independent method for EMG-based myoelectric prosthetic hand," *IEEE Sensors J.*, vol. 21, no. 5, pp. 6623–6633, Mar. 2021.

[4] J. M. Hahne, M. A. Schweisfurth, M. Koppe, and D. Farina, "Simultaneous control of multiple functions of bionic hand prostheses: Performance and robustness in end users," *Sci. Robot.*, vol. 3, no. 19, Jun. 2018, Art. no. eaat3630.

[5] Y. Huang, K. B. Englehart, B. Hudgins, and A. D. C. Chan, "A Gaussian mixture model based classification scheme for myoelectric control of powered upper limb prostheses," *IEEE Trans. Biomed. Eng.*, vol. 52, no. 11, pp. 1801–1811, Nov. 2005.

[6] K. Englehart and B. Hudgins, "A robust, real-time control scheme for multifunction myoelectric control," *IEEE Trans. Biomed. Eng.*, vol. 50, no. 7, pp. 848–854, Jul. 2003.

[7] R. Song, K. Y. Tong, X. Hu, and L. Li, "Assistive control system using continuous myoelectric signal in robot-aided arm training for patients after stroke," *IEEE Trans. Neural Syst. Rehabil. Eng.*, vol. 16, no. 4, pp. 371–379, Aug. 2008.

[8] H. S. Lo and S. Q. Xie, "Exoskeleton robots for upper-limb rehabilitation: State of the art and future prospects," *Med. Eng. Phys.*, vol. 34, no. 3, pp. 261–268, Apr. 2012.

[9] D. Leonardis *et al.*, "An EMG-controlled robotic hand exoskeleton for bilateral rehabilitation," *IEEE Trans. Haptics*, vol. 8, no. 2, pp. 140–151, Jun. 2015.

[10] E. P. Doheny, C. Goulding, M. W. Flood, L. Mcmanus, and M. M. Lowery, "Feature-based evaluation of a wearable surface EMG sensor against laboratory standard EMG during force-varying and fatiguing contractions," *IEEE Sensors J.*, vol. 20, no. 5, pp. 2757–2765, Mar. 2020.

[11] A. Moin *et al.*, "A wearable biosensing system with in-sensor adaptive machine learning for hand gesture recognition," *Nature Electron.*, vol. 4, no. 1, pp. 54–63, Jan. 2021.

[12] X. Jiang *et al.*, "Enhancing IoT security via cancelable HD-sEMG-based biometric authentication password, encoded by gesture," *IEEE Internet Things J.*, vol. 8, no. 22, pp. 16535–16547, Nov. 2021.

[13] X. Chen and Z. J. Wang, "Pattern recognition of number gestures based on a wireless surface EMG system," *Biomed. Signal Process. Control*, vol. 8, no. 2, pp. 184–192, Mar. 2013.

[14] X. Zhang and P. Zhou, "High-density myoelectric pattern recognition toward improved stroke rehabilitation," *IEEE Trans. Biomed. Eng.*, vol. 59, no. 6, pp. 1649–1657, Jun. 2012.

[15] B. Hudgins, P. Parker, and R. N. Scott, "A new strategy for multifunction myoelectric control," *IEEE Trans. Biomed. Eng.*, vol. 40, no. 1, pp. 82–94, Jan. 1993.

[16] A. Ameri, M. A. Akhaee, E. Scheme, and K. Englehart, "A deep transfer learning approach to reducing the effect of electrode shift in EMG pattern recognition-based control," *IEEE Trans. Neural Syst. Rehabil. Eng.*, vol. 28, no. 2, pp. 370–379, Feb. 2020.

[17] L. Wu, X. Zhang, K. Wang, X. Chen, and X. Chen, "Improved high-density myoelectric pattern recognition control against electrode shift using data augmentation and dilated convolutional neural network," *IEEE Trans. Neural Syst. Rehabil. Eng.*, vol. 28, no. 12, pp. 2637–2646, Dec. 2020.

[18] X. Zhang, L. Wu, B. Yu, X. Chen, and X. Chen, "Adaptive calibration of electrode array shifts enables robust myoelectric control," *IEEE Trans. Biomed. Eng.*, vol. 67, no. 7, pp. 1947–1957, Jul. 2020.

[19] R. Hu, X. Chen, X. Zhang, and X. Chen, "Adaptive electrode calibration method based on muscle core activation regions and its application in myoelectric pattern recognition," *IEEE Trans. Neural Syst. Rehabil. Eng.*, vol. 29, pp. 11–20, 2021.

[20] R. N. Khushaba, A. H. Al-Timemy, A. Al-Ani, and A. Al-Jumaily, "A framework of temporal-spatial descriptors-based feature extraction for improved myoelectric pattern recognition," *IEEE Trans. Neural Syst. Rehabil. Eng.*, vol. 25, no. 10, pp. 1821–1831, Oct. 2017.

[21] J. Cheng, F. Wei, C. Li, Y. Liu, A. Liu, and X. Chen, "Position-independent gesture recognition using sEMG signals via canonical correlation analysis," *Comput. Biol. Med.*, vol. 103, pp. 44–54, Dec. 2018.

[22] J. W. Robertson, E. J. Scheme, and K. B. Englehart, "Effects of confidence-based rejection on usability and error in pattern recognition-based myoelectric control," *IEEE J. Biomed. Health Inform.*, vol. 23, no. 5, pp. 2002–2008, Sep. 2019.

[23] J. Tomczyński, P. Kaczmarek, and T. Mańkowski, "Hand gesture-based interface with multichannel sEMG band enabling unknown gesture discrimination," in *Proc. 10th Int. Workshop Robot Motion Control (RoMoCo)*, Jul. 2015, pp. 52–57.

[24] E. J. Scheme, B. S. Hudgins, and K. B. Englehart, "Confidence-based rejection for improved pattern recognition myoelectric control," *IEEE Trans. Biomed. Eng.*, vol. 60, no. 6, pp. 1563–1570, Jun. 2013.

[25] S. Amsuess *et al.*, "Context-dependent upper limb prosthesis control for natural and robust use," *IEEE Trans. Neural Syst. Rehabil. Eng.*, vol. 24, no. 7, pp. 744–753, Jul. 2016.

[26] Q. Ding, Z. Li, X. Zhao, Y. Xiao, and J. Han, "Real-time myoelectric prosthetic-hand control to reject outlier motion interference using one-class classifier," in *Proc. 32nd Youth Academic Annu. Conf. Chin. Assoc. Autom. (YAC)*, May 2017, pp. 96–101.

[27] L. Wu, X. Zhang, X. Chen, and X. Chen, "Visualized evidences for detecting novelty in myoelectric pattern recognition using 3D convolutional neural networks," in *Proc. 41st Annu. Int. Conf. IEEE Eng. Med. Biol. Soc. (EMBC)*, Jul. 2019, pp. 2641–2644.

[28] Y.-H. Liu and H.-P. Huang, "Towards a high-stability EMG recognition system for prosthesis control: A one-class classification based non-target EMG pattern filtering scheme," in *Proc. IEEE Int. Conf. Syst., Man Cybern.*, Oct. 2009, pp. 4752–4757.

[29] Q. Ding, X. Zhao, J. Han, C. Bu, and C. Wu, "Adaptive hybrid classifier for myoelectric pattern recognition against the interferences of outlier motion, muscle fatigue, and electrode doffing," *IEEE Trans. Neural Syst. Rehabil. Eng.*, vol. 27, no. 5, pp. 1071–1080, May 2019.

[30] E. J. Scheme, K. B. Englehart, and B. S. Hudgins, "Selective classification for improved robustness of myoelectric control under nonideal conditions," *IEEE Trans. Biomed. Eng.*, vol. 58, no. 6, pp. 1698–1705, Jun. 2011.

[31] E. J. Scheme and K. B. Englehart, "Validation of a selective ensemble-based classification scheme for myoelectric control using a three-dimensional Fitts' law test," *IEEE Trans. Neural Syst. Rehabil. Eng.*, vol. 21, no. 4, pp. 616–623, Jul. 2013.

[32] L. Wu, X. Zhang, X. Zhang, X. Chen, and X. Chen, "Metric learning for novel motion rejection in high-density myoelectric pattern recognition," *Knowl.-Based Syst.*, vol. 227, Sep. 2021, Art. no. 107165.

[33] A. Bendale and T. E. Boult, "Towards open set deep networks," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 1563–1572.

[34] H.-M. Yang, X.-Y. Zhang, F. Yin, and C.-L. Liu, "Robust classification with convolutional prototype learning," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 3474–3482.

[35] H.-M. Yang, X.-Y. Zhang, F. Yin, Q. Yang, and C.-L. Liu, "Convolutional prototype network for open set recognition," *IEEE Trans. Pattern Anal. Mach. Intell.*, early access, Dec. 16, 2020, doi: 10.1109/TPAMI.2020.3045079.

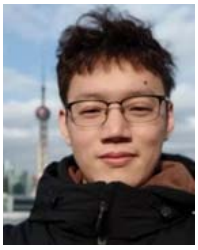
[36] X. Jiang *et al.*, "Open access dataset, toolbox and benchmark processing results of high-density surface electromyogram recordings," *IEEE Trans. Neural Syst. Rehabil. Eng.*, vol. 29, pp. 1035–1046, 2021.

[37] D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," 2014, *arXiv:1412.6980*.

[38] J. A. Hanley and B. J. McNeil, "The meaning and use of the area under a receiver operating characteristic (ROC) curve," *Radiology*, vol. 143, no. 1, pp. 29–36, 1982.

[39] L. Van der Maaten and G. Hinton, "Visualizing data using t-SNE," *J. Mach. Learn. Res.*, vol. 9, no. 11, pp. 2579–2605, 2008.

[40] A. T. Nguyen *et al.*, "A bioelectric neural interface towards intuitive prosthetic control for amputees," *J. Neural Eng.*, vol. 17, no. 6, Nov. 2020, Art. no. 066001.



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